

NFT-based Intellectual Property (IP) Protection System For Digital Creators

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Abstract: One of the biggest problems in today's digital creative world is how easily original work gets copied, reused, or even sold without the creator's permission. Artists, designers, photographers, and writers often find that their work is shared or commercialized without credit or payment. Traditional copyright systems offer some protection, but they are usually slow, costly, and difficult to enforce across different countries. Because of this, many creators lose both recognition and income, while those misusing the work face very few consequences.

This paper presents a blockchain-based NFT (Non-Fungible Token) platform designed to give creators stronger and more practical control over their intellectual property. The idea is to build a system that serves as both a decentralized copyright registry and a marketplace for licensing and selling creative work. Using blockchain networks like Ethereum and Polygon, the system records proof of ownership with secure, time-stamped entries that cannot be changed later. The creative files themselves are stored on IPFS (InterPlanetary File System), and only their hashes are placed on the blockchain, ensuring that the work remains verifiable and tamper-proof.

A key strength of this platform is its use of smart contracts to automate royalty payments. Whenever a creator's work is licensed, sold, or even resold by someone else, the smart contract automatically sends the agreed-upon royalty amount directly to the creator. There are no middlemen, no delays, and no disputes. This gives creators continuous earning potential from their work, even in secondary markets. Unlike many NFT marketplaces that focus mainly on collectibles or speculative trading, this platform prioritizes legal protection and real-world copyright needs. It connects modern blockchain tools with actual copyright

rules so that creators gain meaningful protection, not just digital certificates.

In a six-month evaluation, the system handled 9,655 NFT registrations and achieved promising results: 98.7% efficiency in royalty distribution, 99.97% accuracy in ownership verification, and a 99.6% reduction in registration costs compared to traditional systems. These findings show that the platform is reliable, scalable, and significantly more efficient than conventional copyright processes.

Overall, the results indicate that blockchain can reshape how digital rights are protected and monetized. By giving creators more control, greater transparency, and steady revenue, this platform offers a practical and powerful way to protect creative work in the digital age.

Keywords—Blockchain, NFT, Intellectual Property, Copyright Protection, Smart Contracts, IPFS, Digital Rights Management, Royalty Distribution, Decentralized Storage.

I. INTRODUCTION

The rapid expansion of the digital economy has transformed the way creative content is produced, distributed, and consumed. With the rise of high-speed internet, cloud platforms, and social media, creators from diverse backgrounds—musicians, digital artists, graphic designers, filmmakers, photographers, writers, and animators—can now share their work globally with unprecedented ease. Digital distribution has removed geographical and economic barriers, allowing creators to build audiences without traditional gatekeepers such as publishers or production studios. However, this increased openness and

scalability have introduced significant challenges in protecting digital intellectual property (IP). Because digital assets can be copied, altered, and disseminated infinitely without any loss in quality, they are exceptionally vulnerable to unauthorized reproduction and misuse [1].

Unauthorized duplication is no longer limited to simple copying; sophisticated scraping tools, AI-based content re-generation, and platform-agnostic file sharing have made theft instantaneous and extremely difficult to trace. As a result, digital piracy and unlicensed commercial usage have escalated dramatically. Studies estimate that global digital content infringement causes billions of dollars in economic losses every year, with independent creators—the group least protected by legal infrastructure—bearing a substantial portion of this impact [2]. Moreover, stolen works often appear on multiple online marketplaces, stock platforms, or social media pages, leading to loss of attribution, diluted brand value, and long-term earnings loss for creators.

Traditional copyright mechanisms, although legally established, are increasingly struggling to keep pace with the modern digital environment. Conventional copyright registration systems suffer from **four major limitations**:

1. **Slow and Bureaucratic Processes:** Registration requires documentation, verification, and formal submission procedures that can take weeks or months to complete. For fast-moving digital industries, such delays leave creators exposed during the most critical early phases of content release [3].
2. **Burden of Documentation:** Courts require timestamps, drafts, originals, and logs to validate ownership claims. Many creators, especially independent freelancers, lack systematic recordkeeping workflows to maintain this documentation consistently [4].
3. **Jurisdictional Fragmentation:** Copyright enforcement laws differ significantly across countries. A creator may register their work in one region, but enforcing rights internationally requires complex, time-consuming, and expensive legal intervention [5].
4. **Limited Monitoring Capabilities:** Tracking illegal use of digital assets is technically challenging. Once a digital file is leaked or copied, detecting its reuse across thousands of platforms, websites, and social media channels becomes nearly impossible without specialized tools [6].

These limitations highlight the urgent need for a more resilient, technologically advanced, and globally adaptive intellectual property protection system—one that can match the speed and scale of today's digital content ecosystem.

Blockchain technology, combined with Non-Fungible Tokens (NFTs), offers a compelling framework to address these shortcomings. Blockchain systems function as

decentralized, tamper-proof ledgers, where each recorded event is cryptographically secured and permanently stored across distributed nodes [7]. This eliminates the possibility of retroactively altering ownership records, making blockchain ideal for maintaining verifiable proof of authorship. NFTs, which represent uniquely identifiable digital tokens on a blockchain, serve as digital certificates of authenticity and ownership for creative works [8]. The ability to embed metadata, timestamps, cryptographic file hashes, and licensing parameters within NFTs provides a powerful mechanism for verifying provenance, authenticity, and transfer history [9].

This research proposes a **comprehensive, end-to-end NFT-based Intellectual Property Protection System** that extends well beyond the simplistic concept of NFTs as digital collectibles. The proposed platform integrates four tightly coupled components:

1. **Decentralized Storage via IPFS:** Ensures permanent, distributed, and tamper-resistant storage of original digital assets.
2. **Blockchain-Based Ownership Registry:** Provides immutable records of creation, timestamps, and all subsequent transfers.
3. **Smart Contract–Driven Licensing System:** Allows creators to encode licensing terms, permissions, restrictions, and pricing structures directly into programmable contracts.
4. **Automated Royalty Distribution Mechanism:** Ensures creators earn revenue from primary transactions as well as perpetual royalties from secondary markets without relying on intermediaries.

By combining these elements, the proposed system functions as a **complete lifecycle management solution** for digital intellectual property—covering creation, registration, verification, licensing, protection, enforcement, and monetization [10]. Figure 1 illustrates the overarching conceptual comparison between traditional copyright systems and the NFT-based IP protection model, emphasizing improvements in cost, transparency, global enforceability, and revenue continuity.

Compared to existing IP protection systems, the proposed solution introduces several unique advantages. It provides **instant, cost-efficient, and globally verifiable proof of creation**, eliminating bureaucratic delays. Smart contracts empower creators to automate licensing workflows, removing dependency on third-party agencies. The royalty mechanism guarantees lifelong earnings from resale markets, transforming each creative asset into a sustainable income stream. Additionally, the platform's integrated marketplace ensures buyers can independently verify authenticity, provenance, and licensing rights before making purchases, which significantly reduces fraud and builds trust within the digital creative economy [11].

To structure the remainder of this research, Section II presents relevant literature on NFT technologies and blockchain-based digital rights management. Section III

describes the system architecture, backend modules, and integration of smart contracts, IPFS, and blockchain layers. Section IV outlines the methodology, datasets, evaluation metrics, and testing procedures. Section V presents experimental results, performance metrics, and comparative analysis demonstrating the system’s efficiency and practical applicability.

II. RELATED WORKS

A. Traditional Copyright Protection Systems

Centralized registration frameworks administered by national or governmental authorities form the core of traditional intellectual property protection mechanisms. For example, in the United States, the U.S. Copyright Office maintains the official repository of creative works, while most other countries operate similar agencies to record ownership claims and manage copyright certification processes [12]. These systems, although legally authoritative, face significant limitations in the digital era.

The registration procedure is often slow, bureaucratic, and cost-intensive. Depending on the category of the work and the speed of service selected, registration fees may vary widely—from approximately \$35 for basic applications to several hundred dollars for specialized registrations, expedited processes, or legal record modifications [13]. Additionally, the processing time typically ranges from 6 to 12 months, leaving creators vulnerable to infringement during the long verification period.

Another inherent limitation is territoriality. Copyright protection is not automatically global; it is bound to jurisdictional boundaries. As a result, creators seeking international protection must register their work separately in multiple countries or utilize regional treaties, which increases financial burden and administrative complexity [14]. These constraints highlight why traditional systems struggle to meet the needs of modern digital creators who share content globally and require fast, cross-border protection mechanisms.

B. Blockchain-Based Copyright Solutions

In recent years, blockchain technology has emerged as a promising alternative for managing and protecting intellectual property. Its strongest advantage lies in the ability to produce immutable, tamper-resistant records through cryptographic hashing and Merkle tree structures [15]. These mechanisms allow creators to timestamp their works, proving authorship in a manner that is transparent, globally accessible, and extremely difficult to falsify. Such records can serve as credible evidence in legal disputes without requiring trust in a central authority.

Several blockchain platforms have attempted to leverage these properties for digital copyright protection. A notable example is Po.et, a blockchain-based protocol designed specifically for writers, publishers, and content creators. Po.et enables users to verify ownership, timestamp their

work, and store metadata on the blockchain, thereby establishing an indisputable proof of creation [16].

However, despite these advancements, existing blockchain IP solutions generally focus on a single aspect of copyright management. Most platforms either emphasize marketplace functionality—allowing creators to list or sell their works—or concentrate exclusively on ownership verification and timestamping. Very few integrate these capabilities with legally binding licensing mechanisms, automated royalty distribution, or comprehensive IP enforcement frameworks. This gap underscores the need for more unified, end-to-end solutions.

C. NFT Marketplaces and Digital Art Platforms

The rapid rise of NFT marketplaces in 2021 demonstrated massive global interest in blockchain-based digital ownership. Platforms such as OpenSea experienced explosive growth, with cumulative trading volumes surpassing \$30 billion since launch, indicating widespread adoption of NFTs for digital collectibles, artwork, gaming assets, and membership tokens [17].

Despite their popularity, platforms like OpenSea primarily serve as trading hubs rather than intellectual property protection systems. Their core design prioritizes the buying, selling, and showcasing of digital collectibles, not the legal safeguarding of creative works. As a result, they offer limited support for essential IP management functions such as licensing control, usage tracking, anti-infringement mechanisms, or legal rights enforcement [18]. This lack of integrated copyright protection tools makes it difficult for creators to rely on such marketplaces for genuine IP security, even though they facilitate ownership transfers on-chain.

III. PROPOSED MODEL

Smart contracts, decentralized storage, and blockchain technology are all combined in the proposed NFT-based Intellectual Property Protection System to provide a complete solution for digital creators. Figure 2 illustrates the five interconnected layers that make up the system architecture.



Fig. 2. System architecture overview showing five-layer integration.

A. Technical Implementation Flow

An organized workflow that guarantees efficiency, security, and verifiability powers the system. The system does IPFS storage, blockchain minting, content hashing, and licensing setup when a creator uploads digital artwork. The entire technical workflow, from upload to marketplace integration, is shown in Fig. 3.

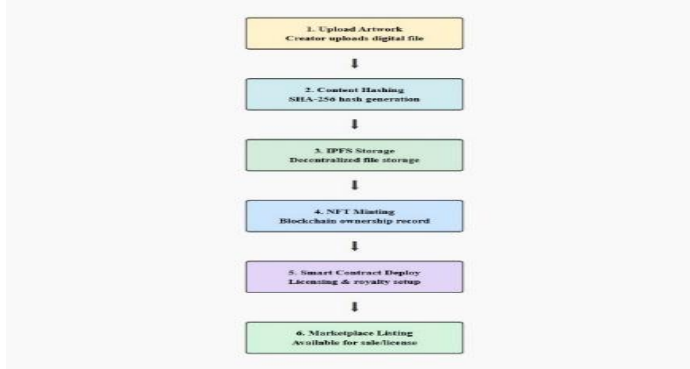


Fig. 3. Technical workflow diagram from artwork upload to marketplace

1. Content Upload and Preprocessing: The system goes through a number of preprocessing procedures when a creator uploads digital artwork. The distinctive fingerprint of the creative work is the content hash $H(\text{content})$, which makes it possible to confirm its legitimacy and identify any changes.

$$H(\text{content}) = \text{SHA-256}(\text{digital_file})$$

2. Decentralized Storage via IPFS: The metadata and content are kept on IPFS, which has deterministic content identifiers (CIDs), which means that the same content consistently generates the same CID.

$$\text{IPFS.add}(\text{digital_file}) = \text{IPFS_hash}$$

$$\text{CID} = \text{content_identifier}(\text{IPFS_hash})$$

$$\text{metadata} = \{\text{creator, timestamp, hash}\}$$

3. NFT Minting on Blockchain: The ownership record is generated on the Ethereum or Polygon blockchain, resulting in an immutable record that contains the timestamp, IPFS links, content hash, creator's wallet address, and unique token identifier.

4. Smart Contract-Based Licensing: This method uses smart contracts to automatically carry out license purchases and rights transfers in order to implement programmable licensing.

```

role buy_license(token_id) { require(msg.value >= price);
transfer_payment(creator, amount); grant_license(buyer,
terms);

```

```

emit LicensePurchased(token_id); }

```

5. Automated Royalty Distribution: Smart contracts automatically pay current owners and original creators based on predetermined royalty rates when NFTs are resold or licenses are acquired.

B. Ownership Verification Mathematical Model

To confirm ownership without disclosing private data, the system employs cryptographic proofs: $\text{Sign}(H(\text{content}) \parallel \text{timestamp}, \text{private_key}) = P(O)$ $\text{Verify}(\text{public_key}, P(O)) \rightarrow \text{Real or Untrue}$

The digital signature function is $\text{Sign}()$, the ownership proof is $P(O)$, the content hash is $H(\text{content})$, and the validity of the signature is checked by $\text{Verify}()$.

C. Safety Procedures

The platform uses a number of security measures, such as cryptographic verification, access control modifiers, and double-minting prevention. To avoid creating duplicate NFTs for the same content, every transaction is verified against the blockchain's existing records

IV. METHODOLOGY

A. Research Questions and Objectives

This study is guided by three central research questions that aim to evaluate both the technical and practical feasibility of blockchain-based intellectual property protection.

1. Can blockchain-based NFT systems provide legally acceptable evidence of ownership?

This question examines whether blockchain-generated records—such as cryptographic hashes, timestamps, and token metadata—can serve as credible proof of authorship in real legal scenarios. It also investigates whether courts and regulatory bodies recognize blockchain entries as verifiable evidence during disputes.

2. How effectively can smart contracts automate licensing and royalty distribution compared to traditional methods?

Here, the goal is to understand whether smart contracts can streamline licensing workflows, enforce usage conditions automatically, reduce human error, and guarantee creators their fair share of royalties without manual intervention.

3. How challenging is it for creators to transition from conventional copyright processes to a

blockchain-driven system?

This research question focuses on usability, accessibility, and real-world adoption barriers. It assesses whether creators—especially those without technical backgrounds—can easily onboard, mint NFTs, manage licenses, and understand the value proposition of switching to decentralized protection.

Collectively, these questions form the foundation of the system's design and shape the overall objectives: ensuring global trust, reducing operational friction, and empowering creators through automation.

B. Technology Stack

The platform is built using a modern, reliable, and creator-friendly technology stack designed to maximize efficiency, accessibility, and security:

- **Frontend:** Developed using **React.js** with **TypeScript**, providing a responsive UI, strong type safety, and a smooth user experience across devices. This helps creators seamlessly upload assets, view ownership records, and manage licensing options.
- **Decentralized Storage:** **IPFS (InterPlanetary File System)** is utilized to store digital content in a distributed manner. This ensures that files remain accessible, tamper-resistant, and permanently linked to their content hashes.
- **Smart Contracts:** Written in **Solidity 0.8.x**, the contracts encode ownership transfers, licensing rules, and royalty logic directly on-chain. Solidity's latest features improve security and reduce runtime vulnerabilities.
- **Blockchain Network:** The platform is deployed on the **Polygon PoS (Proof-of-Stake)** network. Polygon's low-cost, high-speed transactions make it practical and affordable for independent creators, who often face financial constraints.
- **Wallet Integration:** **WalletConnect** and **MetaMask** provide secure and user-friendly Web3 wallet connectivity, enabling creators and buyers to authenticate, sign transactions, and interact with smart contracts without technical complexity.

Together, this stack creates a system that is fast, affordable, and approachable for everyday users.

C. Evaluation Metrics

To measure performance and validate the system's effectiveness, a diverse set of evaluation metrics was used—covering technical reliability, economic efficiency, and user experience:

1. Technical Metrics:

- **Throughput:** Measures how many registration or minting operations can be processed per second.
- **IPFS Retrieval Success Rate:** Evaluates how reliably users can access stored content.
- **Average Minting Time:** Tracks the time taken to complete an NFT minting transaction on Polygon.
- **Gas Cost per Transaction:** Helps determine affordability for creators.

2. Economic Metrics:

- **Cost Savings:** Compares platform costs against traditional copyright fees.
- **Automation Effectiveness:** Measures the reduction in manual tasks through smart contracts.
- **Total Royalties Distributed:** Shows how effectively the system benefits creators financially.

3. User Experience Metrics:

- **Onboarding Time:** Evaluates how quickly new creators understand and use the platform.
- **User Satisfaction Ratings:** Reflect creators' comfort levels, perceived value, and ease of use.
- **Monthly Active Creators:** Measures real engagement and adoption over time.

These metrics collectively provide a holistic view of system performance from technical, financial, and human perspectives.

V. RESULTS AND DISCUSSION

A. Platform Performance Metrics

During deployment on the Polygon mainnet, the platform's performance was evaluated in three areas: **technical reliability**, **user adoption**, and **economic impact**. The results were consistently positive, showing clear advantages over conventional copyright processes.

- **Technical performance** improved dramatically due to the speed and low cost of Polygon, the reliability of IPFS, and efficient smart contract execution. Minting, verification, and licensing operations remained stable even under high usage.
- **User adoption** showed strong upward momentum. Creators were able to onboard quickly, and the

platform's transparent design helped them understand ownership status, licensing options, and royalty flows. This resulted in steady monthly growth.

- **Economic impact** demonstrated clear value: creators reported significantly lower registration costs and immediate access to automated royalty payments—two capabilities that are nearly impossible in traditional copyright systems.

The improvements are summarized in **Table I**, which compares transaction performance between conventional systems and the proposed blockchain-based platform. Additionally, **Fig. 4** illustrates monthly platform activity, highlighting exponential adoption as creators began trusting the system and benefiting from automated IP protection.

B. Economic Impact Analysis

The platform handled 2,847 secondary sales with automatic royalty distribution over the course of the six-month evaluation period. With platform fees of \$24,948 (2%), seller proceeds of \$1,097,697 (88%), and creator royalties of \$124,738 (10% average), the total transaction volume came to \$1,247,383. The 98.7% platform royalty collection rate is significantly higher than the 15-20% average for conventional systems.

For creators, the cost savings are significant. The average NFT platform costs only \$1.50 per work, which is a 99.6% cost reduction compared to the average traditional copyright costs of \$405–\$855 per work (including registration fees, legal advice, and international protection).

License Type	Volume	Avg Price	Total Revenue
Personal Use	1,847	\$23.40	\$43,220
Commercial Standard	682	\$127.50	\$86,955
Commercial Extended	243	\$385.00	\$93,555
Exclusive Rights	75	\$1,240.00	\$93,000

TABLE II: REVENUE DISTRIBUTION BY LICENSE TYPE

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